Differentiation Rules: Power, Product, Quotient, Chain

Power, sum, and constant rules

1 Differentiate each function:

a $f(x) = 5x^4 - 3x^2 + 7$ **b** $f(x) = \frac{1}{x^3}$ **c** $f(x) = 4\sqrt{x}$

Product and quotient rules

2 Differentiate $f(x) = (2x+1)(x^2-3)$ using the product rule.

3 Differentiate $f(x) = \frac{x^2+1}{x-2}$ using the quotient rule.

4 Differentiate $f(x) = x^2 \sin x$.

The chain rule

5 Differentiate each function using the chain rule:

a $f(x) = (3x^2-5)^4$ **b** $f(x) = \sin(2x^3)$

c $f(x) = e^{-x^2}$ **d** $f(x) = \ln(x^2+1)$

6 Differentiate $f(x) = \sqrt{4x + \cos x}$.

Combined rules

7 Differentiate $f(x) = \frac{\sin x}{x^2}$ using the quotient rule.

8 Differentiate $f(x) = e^{2x}\cos(3x)$ using the product and chain rules together.

9 Differentiate $f(x) = \left(\frac{x+1}{x-1}\right)^3$ using the chain rule together with the quotient rule.

Tangent lines using derivatives

10 Find the equation of the tangent line to $f(x) = x^2 - 3x + 1$ at $x = 2$.

11 Find the point(s) on the curve $y = x^3 - 3x$ where the tangent line is horizontal.

Higher-order derivatives

12 Find $f''(x)$ for $f(x) = x^4 - 6x^2 + 2x$.

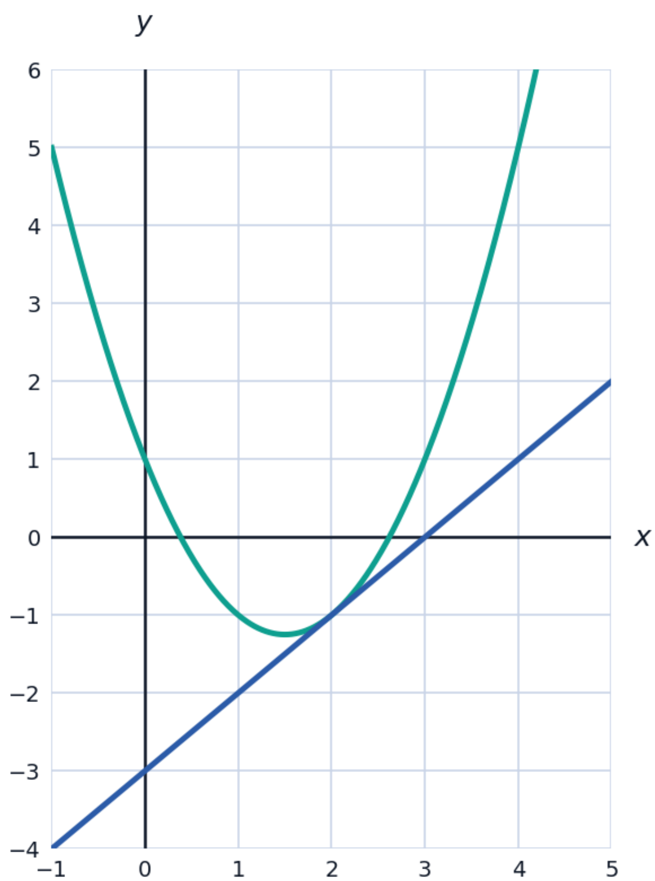
13 Find $f''(x)$ for $f(x) = \sin(2x)$.

Normal lines

14 Find the equation of the line normal (perpendicular) to $f(x) = x^2$ at $x = 2$.

Visualizing a function and its tangent line

15 The function $f(x) = x^2 - 3x + 1$ and its tangent line at $x = 2$ (found earlier, $y = x - 3$) are graphed together. Use the graph to confirm the tangent line touches the curve at exactly the point $(2, -1)$.



Differentiating inverse trig and more transcendental functions

16 Differentiate each function:

a $f(x) = \tan^{-1}(3x)$

b $f(x) = \sin^{-1}(x^2)$

c $f(x) = x \ln x$

Implicit differentiation (a preview)

17 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$ by differentiating both sides with respect to x (treating y as a function of x).

Related applications of the derivative rules

18 A particle's position is $s(t) = t^3 - 6t^2 + 9t$. Find its velocity function $v(t) = s'(t)$ and determine when the particle is at rest ($v(t) = 0$).